INCH-POUND

MIL-PRF-47257D
18 February 1998
SUPERSEDING
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25 March 1991

PERFORMANCE SPECIFICATION

COMPOUND, EPOXY, FILAMENT WINDING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a fiberglass roving impregnated with an epoxy resin-hardener compound used in the fabrication of filament wound structures.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Aviation and Missile Command, ATTN: AMSAM-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A AREA CMPS

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1475	-	Density of Paint, Varnish, Lacquer, and Related Products, Test Methods for
ASTM D 2291	-	Fabrication of Ring Test Specimens for Glass-Resin Composites, Standard Practice for
ASTM D 2393	-	Viscosity of Epoxy Resins and Related Components, Standard Test Method for
ASTM D 2471	-	Gel Time and Peak Exothermic Temperature of Reacting Thermosetting Resins, Standard Test Method for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials shall consist of a) an epoxy resin, b) an anhydride hardener, c) an amine promoter, and d) fiberglass roving. Ingredients a, b, and c shall provide not more than 26 ± 4 percent of the product content by weight. (See 4.6.4.1 and 4.6.4.2).

3.2.1 Condition in container. The component materials in containers shall be free of gel particles and foreign matter that would adversely affect their intended purpose. Crystallization shall be eliminated by heating to not greater than 49 °C and stirring shall be acceptable (see 4.6.4). The resin content shall be marked on the container.

3.2.2 Storage life. The individual components shall be capable of meeting the requirements of this specification after storage in the original unopened containers at temperatures between 4 and 32° C in no sunlight for not less than 12 months from the date of shipment (see 4.6.4.3).

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3.2.3 Properties of the mixed compound. The mixed epoxy compound consisting of resin, hardener, and promoter and mixed to the manufacturer's specifications shall exhibit the following properties:

3.2.3.1 Viscosity. The viscosity, immediately after mixing, shall be within the range of 500 to 3000 milli pascal seconds (mPa·s) at 25 ± 1 °C (see 4.6.4.5).

3.2.3.2 Gel time. The mixed compound shall gel to a non-flowable solid in not greater than 1 hour at 149 ± 3 °C (see 4.6.4.6).

3.2.3.3 Product. The impregnated roving product shall be evenly wound on a tube with negligible free liquid compound present. The material shall be readily removable from the container (aluminum alloy foil or polyethylene bag) with negligible adhesion to the container or to the roll and no deterioration of the roving during removal (see 4.6.4).

3.3 Performance characteristics.

3.3.1 Product. The product shall be capable of meeting the requirements of this specification after storage at a temperature not greater than 10 °C for not less than 30 days, or after storage at room temperature (16 to 32 °C) for not less than 5 days (see 4.6.4.3).

3.3.2 Shear strength. After a cure time of not greater than 3 hours at 141 ± 3 °C, the shear strength of the material shall be not less than 5000 pounds per square inch (psi) at an ambient temperature of 25 ± 1 °C (see 4.6.4.7).

3.4 Warning label. Containers of the epoxy compound shall be marked with a warning label addressing hazards and cautions as identified by the component manufacturers.

3.5 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.3, 4.4, and 4.6.

4.3 First article inspection. Unless otherwise specified in the contractual document (see 6.2), a sample of each type of material, manufactured under the same conditions as those proposed

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for subsequent production, shall be subjected to first article inspection. The sample shall consist of a 1-quart kit containing all the component materials, or a 1-quart sample of each material except the amine promoter, which may be furnished as a 1-pint sample. The sample shall also consist of not less than 5 pounds of impregnated roving. First article inspection shall be as specified herein. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained. Units subjected to first article inspection shall have successfully passed the quality conformance inspection. Testing of the first article sample to determine compliance with the characteristics listed in table I shall be conducted in accordance with the corresponding test paragraphs.

TABLE I. First article inspection.

Characteristics	Test paragraph
Storage life	4.6.4.3
Component materials	
Specific gravity of epoxy resin	4.6.4.4
Specific gravity of anhydride hardener	4.6.4.4
Specific gravity of amine promoter	4.6.4.4
Viscosity of epoxy resin	4.6.4.5
Viscosity of anhydride hardener	4.6.4.5
Mixed compound	
Shear	4.6.4.7

4.4 Conformance inspection.

4.4.1 Lot formation.

4.4.1.1 Component material. A lot shall consist of all the component material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.4.1.2. Compounded material. A lot shall consist of all the compound material mixed in a continuous operation by the same process by the same compounder using the same specified lot of component materials in accordance with this specification and submitted for inspection at one time.

4.4.2 Sampling for examination. Each container of component material shall be considered as a unit of product. Random selection of a container from each lot of component material shall be effected. From these sample component materials, sufficient quantities shall be randomly appropriated for the manufacture of sufficient fiberglass roving materials for subjection to the acceptance inspection as delineated in table II. Failure of this sample to comply with the requirements specified in table II shall be cause for lot rejection.

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TABLE II. Conformance inspection.

Characteristics	Requirement paragraph	Test paragraph
Composition	3.2	4.6.4
Condition in container	3.2.1	4.6.4
Mixed compound		
Viscosity	3.2.3.1	4.6.4.5
Gel Time	3.2.3.2	4.6.4.6

4.5 Inspection equipment. Inspection equipment for conducting examination and tests shall be as specified in the following test methods and procedures paragraphs.

4.6 Test methods and procedures.

4.6.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements:

a. Temperature, room ambient (16 to 32 °C)
b. Altitude, facility ground
c. Humidity, facility ambient up to 95 percent relative humidity.

4.6.2 Test sequence. Test sequence within each classification of inspection (first article or quality conformance) shall be at the option of the contractor.

4.6.3 Test specimen preparation. Unless otherwise specified herein, specimens shall be prepared as follows:

4.6.3.1 Packaged product material. To prevent moisture condensation on the impregnated roving, all packages of material which are removed from the storage cooler (see 3.3.1) shall be allowed to warm to room temperature for a minimum of 2 hours prior to opening the sealed bag.

4.6.3.2 Shear specimens. The filament wound rings shall be prepared from the product material in accordance with ASTM D 2291. If supplied as type I material, the components shall be mixed and type II material prepared. The filament wound rings shall be cured for not greater than 3 hours at 141 $\pm$ 3 °C. Shear specimens shall be fabricated from the rings to the dimensions shown on figure 1.

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NOTE

Observe test precautions when handling the component or compounded materials during the following tests.

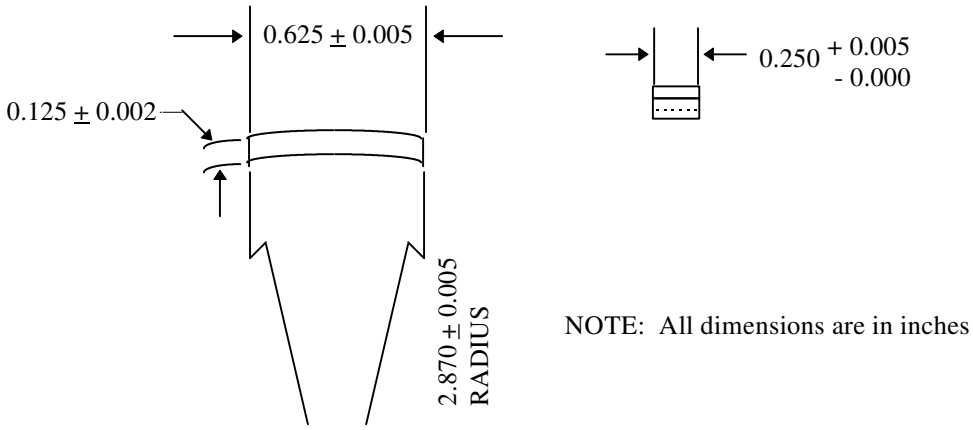


FIGURE 1. Shear test specimen.

4.6.4 Composition. The condition in containers shall be examined visually and shall conform to requirements of 3.2.1. The composition shall be certified by the contractor and shall conform to requirements of 3.2.2.

4.6.4.1 Resin content. The preferred procedure for determining the resin content of the product material shall be as follows:

a. Weigh the empty impregnated roving spool (W_1) prior to installing on the roving impregnating machine.
b. Weigh the dry spool of roving (W_2).
c. After impregnating the roving, weigh the impregnated spool of roving (W_3), and the dry roving spool remainder (W_4)
d. Calculate the resin content by the following formula:

$$R = 100 - 100 \times \frac{W_2 - W_4}{W_3 - W_1}$$

Where R = resin percent content

W_1 = weight of empty impregnated roving spool

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W_2 = weight of dry roving spool

W_3 = weight of impregnated roving spool

W_4 = weight of dry roving spool remainder

Therefore, $W_2 - W_4$ = weight of dry roving

$W_3 - W_1$ = weight of impregnated roving

e. Record the impregnated roving spool number, the net weight of the impregnated roving, the date, the resin formula number, the roving type, and percent resin content on a form affixed on the core of the impregnated roving spool.

4.6.4.2 Resin content, alternate method. The alternate procedure for determining the resin content of the product material shall be as follows:

a. Weigh the specimen on an analytical balance (W_5).
b. Burn the specimen in a muffle furnace at 593 to 621 °C for 1 to 2 hours or until the specimen is white.
c. Cool the specimen in a desiccator.
d. Weigh the specimen again on an analytical balance (W_6).
e. Calculate the resin content as follows:

$$R = \frac{W_5 - W_6}{W_5} \times 100$$

Where R = percent of resin content

W_5 = weight of specimen before burning

W_6 = weight of specimen after burning

4.6.4.3 Storage life. The contractor's certification of compliance to the storage life requirements (see 3.2.2) is required.

4.6.4.4 Specific gravity. The specific gravities of the epoxy resin, anhydride hardener, and amine promoter shall be determined in accordance with ASTM D 1475.

4.6.4.5 Viscosity. The viscosities of the epoxy resin, the anhydride hardener, and the mixed compound shall be determined in accordance with ASTM D 2393 or 3.2.3.1, as applicable.

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4.6.4.6 Gel time. The gel time shall be determined in accordance with ASTM D 2471 and shall conform to 3.2.3.2.

4.6.4.7 Shear strength. The shear strength of test specimens prepared as specified in 4.6.3.2 shall be determined as follows and shall meet the requirements of 3.3.2.

4.6.4.7.1 Equipment. The following equipment shall be used:

a. Compression testing machine
b. Horizontal shear fixture, as shown on figure 2
c. Micrometer.

4.6.4.7.2 Test procedure. The test for shear strength shall be as specified below:

a. Measure the thickness and width of the specimen to the nearest 0.001 inch at midpoint.
b. Place the specimen in test fixture, convex surface up. Align the specimen so that its midpoint is centered under the loading nose.
c. Apply the load at a crosshead speed of 0.05 inches per minute.
d. Record the load required to break the specimen. (A typical shear failure is characterized by a sharp, audible report.)
e. A minimum of three specimens shall be tested from each ring.
f. Values for properties at break shall not be calculated for any specimen that breaks at some obvious flaw, unless such flaw constitutes a variable being studied. Retest shall be made for any specimen which fails in any manner other than shear.
g. Calculate the horizontal shear strength as follows:

$$S_H = \frac{0.75\ P_B}{bd}$$

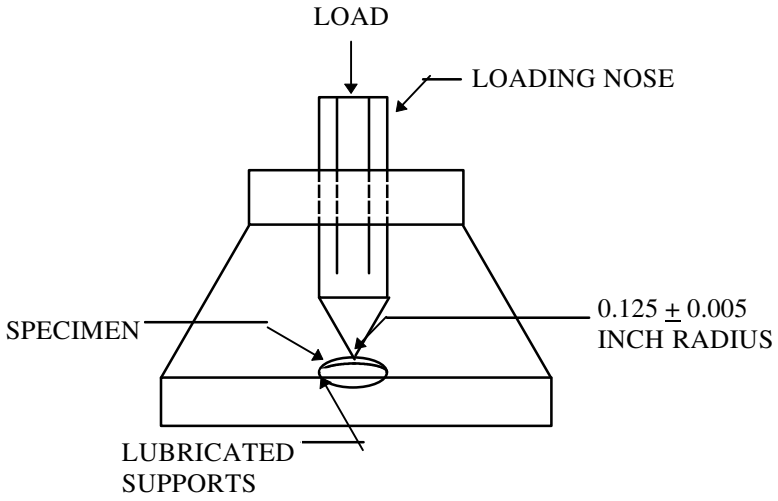
Where: S_H = Apparent horizontal shear strength (psi)

P_B = Breaking load (pounds)

b = Width (inches)

d = Thickness (inches)

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NOTE: Dimension is in inches.

FIGURE 2. Typical horizontal shear fixture.

4.6.4.7.3 Test measurements. The following test measurements shall be recorded:

a. Date of test
b. Material identification (cure, fabrication technique, etc.)
c. Ultimate breaking load
d. Horizontal shear strength
e. Specimen dimensions
f. Test temperature
g. Rate of loading
h. Conditioning (if any)
i. Resin content
j. Specific gravity.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. This material is intended for use as an impregnant for fiberglass filaments used in missile systems. The cured parts are satisfactory for use at temperatures from -54 to 121 °C.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
c. Whether first article is required (see 3.1 and 4.3).
d. Conditions for quality conformance inspection (see 4.4).
e. First article sample size (see 4.3).
f. Packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.6 Subject term (keyword) listing.

Anhydride hardener
Resin-hardener
Amine promoter

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
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Preparing Activity:
Army - MI

Review activities:
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INSTRUCTIONS

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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE	1. DOCUMENT NUMBER MIL-PRF-47257D	2. DOCUMENT DATE (YYMMDD) 18 February 1998

3. DOCUMENT TITLE
COMPOUND, EPOXY, FILAMENT WINDING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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c. ADDRESS (Include Zip Code) U.S. Army Aviation and Missile Command ATTN: AMSAM-RD-SE-TD-ST Redstone Arsenal, AL 35898-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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